

About This Book

This book has 14 chapters:

Part I: Understanding Recursion

Chapter 1: What Is Recursion? Explains recursion and how it is the natural result of the way programming languages implement functions and function calls. This chapter also argues that recursion isn't nearly the elegant, mystical concept many claim it is.

Chapter 2: Recursion vs. Iteration Dives into the differences (and many similarities) between recursive and iterative techniques.

Chapter 3: Classic Recursion Algorithms Covers famous recursive programs such as the Tower of Hanoi, the flood fill algorithm, and others.

Chapter 4: Backtracking and Tree Traversal Algorithms Discusses a problem for which recursion is particularly suited: traversing tree data structures, such as when solving mazes and navigating a directory.

Chapter 5: Divide-and-Conquer Algorithms Discusses how recursion is useful for splitting large problems into smaller subproblems and covers several common divide-and-conquer algorithms.

Chapter 6: Permutations and Combinations Covers recursive algorithms involving ordering and matching, as well as the common programming problems to which these techniques are applied.

Chapter 7: Memoization and Dynamic Programming Explains some simple tricks to improve code efficiency when applying recursion in the real world.

Chapter 8: Tail Call Optimization Covers tail call optimization, a common technique used to improve the performance of recursive algorithms, and how it works.